



Summary

This project will analyze the industrialization and sustainable production of products derived from one of the most abundant native species in the Amazon: *Anacardium occidentale* L. known as "Marañón" and from which the cashew nut comes. Despite being native to the Americas, world production is led by Africa (Ivory Coast) and India. Peruvian varieties have a high content of vitamin C and polyphenols, and antioxidant properties. The cashew is versatile: both its seed (cashew), its fruit, the leaves and the bark have different medicinal uses and can be widely used. This is also considering the growing consumption of organic foods and with substances beneficial to health. The processing and commercialization of cashews could improve the hitherto disadvantaged economy of native communities, who would also contribute with ecological and sustainable practices in its cultivation. Technological innovations are essential for the future of communities, by integrating these tools, not only production is improved, but sustainability and community development are also strengthened.

Introduction

In Peru there is a great diversity of plant species that have been used since our ancestors for healing and nutritional purposes; this particularly in tropical regions that are favored by the proportion of medicinal species known so far and that remain to be explored. These regions have however been excluded from the economic prosperity observed in other regions, and rather subjected to the negative impacts of extractive activities in their territories. The purpose of this project is to show the viability of producing value-added goods from a native species, the cashew, in a sustainable, environmentally friendly way and with a great economic and social impact.

The cashew tree grows widely in Peruvian regions such as Loreto, San Martin, Piura, Junín, Ucayali and Lambayeque, and can be used in its entirety: its seed, fruit, leaves and bark provide excellent medicinal and nutritional properties. The trunk is used for furniture construction; bark for the production of inks; the apple or fruit is used in the production of sweets, preserves, jams, jellies and wines; it is antiscorbutic and diuretic; The nut is used to harden chocolates and in the manufacture of sweets and juices. Two oils are produced in the walnut: one of the shell (pericarp, testa), black, viscous, irritating and caustic that contains phenolic compounds, cardol with applications in the plastics, insecticides and inks industry; The other oil is extracted from the almond, it is bright yellow. The cashew is a vital crop for many native communities. Its production not only provides income, but also fosters local culture and employment. Knowing their value is essential to implementing appropriate technologies.

Methodology and materials

This study will be carried out following a methodology that consists of:
Exploratory analysis of the optimal conditions for cashew production, identifying current production areas in Peru and type of product currently marketed. To this end, an extensive literature review will be carried out.

Identification of communities surrounding these areas that could benefit from their production, and of the relevant socioeconomic aspects of such communities.

1. Environmental sustainability analysis of cashew crops: for example, it is known that cashew can develop in places where there are long intermediate droughts, in addition, in a wide variety of soils (clay, deep sandy and degraded), although it prefers moist conditions and grows best in well-drained soils (Cárdenas and Paye. 2021).

2. Analysis of the process of industrialization of the cashew that involves evaluating the technology, and a detailed cost analysis will be included, fundamental to evaluate the profitability of the industrialization process. Production, distribution, and marketing costs will be included, allowing informed decisions to be made about investments and business strategies.

3. The cashew production process from harvest to packaging must be optimized to ensure the quality of the product and the sustainability of the process, minimizing the environmental impact.



Contextualization of the problem

- The cashew, a wild tree with low branching and an average height of between 5 and 14 meters, is native to tropical areas of the Americas and northwestern Brazil (Mendez, 2014), but is scarcely exploited in the Peruvian Amazon. There are almost no companies producing products derived from cashew almonds. The cashew is a nutritious and healthy food: Its fruit contains 47% fat, 21% protein and 22% carbohydrates, vitamins and amino acids. The fruit is attached to a pseudo fruit or gray nut, which consists of 46.3% oil with respect to its total weight (Méndez, 2014).
- It is important to contribute to the innovation of consumer products, as well as the greater use of crops of native Amazonian species, which are wasted due to lack of information. All this without harming the environment, and at the same time providing the opportunity for the economic improvement of the native inhabitants of the Amazon. Its sustainable cultivation ensures that native communities can maintain their traditions while generating income responsibly.
- **Uses and applications of cashew nuts in the community (Cárdenas & Paye, 2021)**
- The bark and leaves of the cashew are used for the treatment of stomach cramps, inflammations, insomnia, diarrhea, malaria and hemorrhoids.
- The bark and root are also used as a sedative and hypotensive remedy.
- Cashew resin is used to heal skin lesions.
- It is also said to have antimicrobial activity against *S. aureus* and 5 other Gram-positive bacteria.



Discussion of technological solutions

- The production and processing of cashew crops without chemical elements eliminates important sources of environmental pollution such as pesticides and chemical fertilizers. Also, the permanent nature (cashew) of these crops provides environmental services such as soil protection, water filtration and carbon dioxide sequestration in the case of cashew trees. A crucial aspect for the viability of the commercialization of cashew products is the transport that allows a faster and more efficient distribution, reducing the environmental impact and improving connectivity. Therefore, establishing effective distribution networks is vital to connect native communities to markets. Collaboration between producers and distributors can improve access and ensure that products arrive at their destination in optimal condition. For example, you can think about:
- **Use of drones in agriculture:** The use of drones in agriculture allows aerial monitoring of crops. This helps to identify problems such as pests or diseases in time, optimizing the harvest and ensuring a better yield.
- **Agricultural Training and Education:** New agricultural techniques are essential for the success of innovations. Training in the use of technologies is essential. Native communities should be educated on how to use modern tools to maximize their benefits and ensure successful adoption.
- **Collaboration and Networking:** Fostering collaboration between indigenous communities and external organizations can facilitate access to resources and technologies. Support networks are key to implementing effective and sustainable agricultural innovations.
- **Irrigation technology:** Modern irrigation technologies, such as drip irrigation, allow for more efficient use of water. This is crucial in regions where access to water resources is limited, thus improving cashew productivity.

Preliminary conclusions and future work

It has been identified that the tree begins to produce in the second year of planting, considering an average of 200 kg/ha of walnuts and 1,200 kg/ha of false fruit per hectare. By the 8th year, 3 tons/ha of walnuts and 24 tons/ha of false fruit can be produced. All factors are favorable (soil, climate, superior grafting trees, nutrition, irrigation and pest control), extremely high yields can be achieved, with a good quality harvest.

The future of cashew is promising, with an increase in global demand. Continuous innovation in technology and processes will be key to maintaining competitiveness and ensuring the sustainability of the sector in the coming years.

Referencias

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